

KSB MIL CONTROLS LIMITED

USER REQUIREMENTS SPECIFICATION | URS-CV-AUTO-001 | Rev 0

ANNEXURE C

Tag wise Enquiry Data Sheet*(File Format: PDF)*

Role	INPUT — Reference Document
Applicable Parts	Part 2
Description	Tag-wise enquiry datasheets with process data samples for each fluid category — Liquid, Gas, and Steam — for better understanding of sizing variations.

GENERAL	1	Tag No.	PID		1803-FV -10401		2047-1803-EPR-PID-0104-01																												
	2	Service																																	
	3	Line Number	Equipment Name		3"-CL-1803-10407-A1A-IS																														
	4	Area Classification			Zone 1 Gr.IIC T3																														
	5	Ambient Temperature:	Min.	Max.	4.4 °C		46.7 °C																												
	6	Allowable Sound Pressure Level dBA			85 dBA																														
	7	Tightness Requirements			ANSI IV (standard)																														
	8	Available Air Supply Pressure:	Min.	Max.	3.5 kgf/cm²		7 kgf/cm²																												
	9	Power Failure Position			Open																														
	10	Actuator Design Pressure:			kgf/cm²-g																														
PIPE LINE	11	Line Size and Schedule	Inlet	Outlet	3 in STD		3 in STD																												
	12	Pipe Material			Carbon Steel																														
	13	Pipe Insulation			Safety Insulation																														
PROCESS CONDITIONS	14	Process Fluid			SUSPECT LP CONDENSATE																														
	15	Upstream Condition			Water																														
	16	Differential Pressure			12 kgf/cm²																														
	17				Units	@ Max. Flow	@ Norm. Flow	@ Min. Flow																											
	18	Flow Rate			Am³/h	10,5	9,5	4,8																											
	19	Inlet Pressure			kgf/cm²-g	6,8	6,8	6,9																											
	20	Pressure Drop			kgf/cm²	6,57	6,6	6,7																											
	21	Inlet Temperature			°C	99,9	99,9	99,99																											
	22	Inlet Density / Specific Gravity / Molecular Mass			kg/m³	955	955	955																											
	23	Inlet Compressibility Factor			-																														
CALCULATED RESULTS	24	Inlet Viscosity			cP	0,276	0,276	0,276																											
	25	Inlet Specific Heats Ratio			-																														
	26	Inlet Vapour Pressure			kgf/cm²-a	1,031	1,031	1,035																											
	27	Design Pressure / Temperature			12	kgf/cm²-g	150	°C																											
	28	Flow Coefficient Cv			-	5,26	4,7	2,17																											
	29	Travel			%																														
	30	Sound Pressure Level @ Maximum Flow:			dBA	59,	55,4	55,4																											
	BODY AND TRIM	31	MFR	Model	VTA	VTA	POSITIONER	56	MFR	Model	VTA	VTA																							
		32	Body Type		Single Seat Globe			57	Signal: Inlet	Outlet	4-20 mA+HAR	3-15 Psi																							
		33	Body Size	Trim Size	1,5 in	By Vendor		58	Increase Signal Valve: Close																										
34		Rated Cv	Characteris.	By Vendor	Linear	59		Cam Characteristic Direct																											
35		End Connec. & Rating		1,5" 300# RF		60		Bypass	Gauges	No	Yes																								
ACTUATOR		36	Body Material		F.S.ASTM A105 Normalized		61	Position Transmitter		Yes. 4-20mA+Hart																									
		37	Bonnet Type	Material	Bolted Flange	A105	62	Voltage		24 VDC																									
		38	Flow Direction		FTC (Flow-To-Close)		SOLENOID VALVE	63	MFR	Model	-	-																							
		39	Flow Action To		Open			64	Type		-																								
		40	Lubricator	Isolat. Valve	No	No		65	When De-Enegr.Valve:		-																								
	41	Guiding	No. of Ports	By Vendor	1	66																													
	42	Trim Type		Low Noise Trim		SWITCHES		67	MFR	Model	-	-																							
	43	Rated Travel		By Vendor			68	Type	Quantity	-	-																								
	44	Plug/ Ball/ Disk Material		410 (13Cr)			69	Contacts / Rating		-																									
	45	Seat Material		St Gr6 (CoCrAlloy)			70	Switching Position		-																									
46	Cage	Stem Material	-	410 (13Cr)	71																														
ACTUATOR	47	Gasket Material		SPW AISI 316/Graphite		AIR SET	72	MFR	Model	VTA	VTA																								
	48	Stud Bolt Materials		A193 Gr B7/A194 Gr 2H			73	Set Pressure		By Vendor																									
	49	MFR	Model	VTA	VTA		74	Filter	Gauge	Yes	Yes																								
	50	Type		Spring Diaphragm		TESTS	75																												
	51	Size	Area	By Vendor	By Vendor		76	Hydro. Pressure		Yes																									
	52	Air Failure Valve:		Open			77	Leakage		Yes																									
	53	Handwheel		-			78																												
	54	Bench Range		By Vendor			PURCHASE	79	Manufacturer																										
	55					80		Model																											
					81	Purchase Order Num.																													
				82	Price	Item Number																													
				83	Serial Number																														
<table border="1"> <tr> <td>D</td> <td>EFEG</td> <td>29/06/2023</td> <td>IFA-Issue For Approval</td> <td rowspan="4">INSTRUMENT SPECIFICATION Control Valve</td> <td rowspan="4">  </td> </tr> <tr> <td>C</td> <td>ARSM</td> <td>23/02/2023</td> <td>IFA-Issue For Approval</td> </tr> <tr> <td>B</td> <td>ARSM</td> <td>22/12/2022</td> <td>IFA-Issue For Approval</td> </tr> <tr> <td>A</td> <td>JOMM</td> <td>20/06/2022</td> <td>IFA – Issue For Approval</td> </tr> <tr> <td>No.</td> <td>By</td> <td>Date</td> <td>Revision</td> <td>Code: 1</td> <td>Doc. No.: 2047-1803-EIN-DAS-0017</td> <td>Rev.: D</td> </tr> </table>											D	EFEG	29/06/2023	IFA-Issue For Approval	INSTRUMENT SPECIFICATION Control Valve		C	ARSM	23/02/2023	IFA-Issue For Approval	B	ARSM	22/12/2022	IFA-Issue For Approval	A	JOMM	20/06/2022	IFA – Issue For Approval	No.	By	Date	Revision	Code: 1	Doc. No.: 2047-1803-EIN-DAS-0017	Rev.: D
D	EFEG	29/06/2023	IFA-Issue For Approval	INSTRUMENT SPECIFICATION Control Valve																															
C	ARSM	23/02/2023	IFA-Issue For Approval																																
B	ARSM	22/12/2022	IFA-Issue For Approval																																
A	JOMM	20/06/2022	IFA – Issue For Approval																																
No.	By	Date	Revision	Code: 1	Doc. No.: 2047-1803-EIN-DAS-0017	Rev.: D																													

Specification Notes:

1. Full vacuum as design condition. See Process notes.

Process Notes:

1. Additional design condition: FV / 0.5 kg/cm²g, Temperature: 175 °C.
2. Minimum Design Metal Temperature (MDMT): 4.4 °C.
3. Line size to be confirmed based on pump vendor information referenced to minimum flow bypass.
4. Operation Intermittent

				INSTRUMENT SPECIFICATION Control Valve			
D	EFEG	29/06/2023	IFA-Issue For Approval				
C	ARSM	23/02/2023	IFA-Issue For Approval				
B	ARSM	22/12/2022	IFA-Issue For Approval				
A	JOMM	20/06/2022	IFA – Issue For Approval			Sheet 2	of 2
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GENERAL	1	Tag No.	PID		1803-FV -1104		2047-1803-EPR-PID-0011		
	2	Service							
	3	Line Number	Equipment Name		12"-SL-1803-1111-B2A-IH				
	4	Area Classification			Zone 1 Gr.IIC T3				
	5	Ambient Temperature:	Min.	Max.	4.4 °C		46.7 °C		
	6	Allowable Sound Pressure Level dBA			85 dBA				
	7	Tightness Requirements			ANSI IV (standard)				
	8	Available Air Supply Pressure:	Min.	Max.	3.5 kgf/cm²		7 kgf/cm²		
	9	Power Failure Position			Close				
	10	Actuator Design Pressure:			kgf/cm²-g				
PIPE LINE	11	Line Size and Schedule	Inlet	Outlet	12 in XS		12 in XS		
	12	Pipe Material			Carbon Steel				
	13	Pipe Insulation			Hot Insulation				
PROCESS CONDITIONS	14	Process Fluid			LP Steam				
	15	Upstream Condition			Steam				
	16	Differential Pressure			8 kgf/cm²				
	17				Units	@ Max. Flow	@ Norm. Flow	@ Min. Flow	
	18	Flow Rate			kg/h	6405	5823	2912	
	19	Inlet Pressure			kgf/cm²-g	3,31	3,31	3,32	
	20	Pressure Drop			kgf/cm²	0,3	0,31	0,32	
	21	Inlet Temperature			°C	160	160	160	
	22	Inlet Density / Specific Gravity / Molecular Mass			-		18,02		
	23	Inlet Compressibility Factor			-	0,966	0,966	0,966	
CALCULATED RESULTS	24	Inlet Viscosity			cP	0,014	0,014	0,014	
	25	Inlet Specific Heats Ratio			-	1,317	1,317	1,317	
	26	Inlet Vapour Pressure							
	27	Design Pressure / Temperature			8	kgf/cm²-g	220	°C	
	28	Flow Coefficient Cv			-	305	273	134	
	29	Travel			%				
	30	Sound Pressure Level @ Maximum Flow:			dBA	59,8	60,4	60,9	
	BODY AND TRIM	31	MFR	Model	VTA	VTA			
		32	Body Type			Single Seat Globe			
		33	Body Size	Trim Size	8 in	By Vendor			
34		Rated Cv	Characteris.	By Vendor	Equal%				
35		End Connec. & Rating			8" 300# RF				
36		Body Material			C.S.ASTM A216 WCB				
37		Bonnet Type	Material	Bolted Flange	A216 WCB				
38		Flow Direction			FTO (Flow-To-Open)				
39		Flow Action To			Open				
40		Lubricator	Isolat. Valve	No	No				
ACTUATOR	41	Guiding	No. of Ports	By Vendor	1				
	42	Trim Type			Contoured Trim				
	43	Rated Travel			By Vendor				
	44	Plug/ Ball/ Disk Material			410 (13Cr)				
	45	Seat Material			St Gr6 (CoCrAlloy)				
	46	Cage	Stem Material	-	410 (13Cr)				
	47	Gasket Material			SPW AISI 316/Graphite				
	48	Stud Bolt Materials			A193 Gr B7/A194 Gr 2H				
	49	MFR	Model	VTA	VTA				
	50	Type			Spring Diaphragm				
INSTRUMENT SPECIFICATION	51	Size	Area	By Vendor	By Vendor				
	52	Air Failure Valve:			Close				
	53	Handwheel			-				
	54	Bench Range			By Vendor				
	55								
						PURCHASE	79	Manufacturer	
							80	Model	
							81	Purchase Order Num.	
							82	Price	Item Number
							83	Serial Number	
D B A	EFEG	29/06/2023	IFA-Issue For Approval		INSTRUMENT SPECIFICATION Control Valve				
	ARSM	22/12/2022	IFA-Issue For Approval						
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Specification Notes:

1. Valve shall be tested and certified by IBR or IBR authorized representative as per Appendix 'L' of IBR.
2. Full vacuum as design condition. See Process notes.

Process Notes:

1. Additional design condition: FV, Temperature: 220 °C.
2. Minimum Design Metal Temperature (MDMT): 4.4 °C.
3. IBR Applicable.
4. Operation Intermittent

				INSTRUMENT SPECIFICATION Control Valve			
D	EFEG	29/06/2023	IFA-Issue For Approval				
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GENERAL	1	Tag No.	PID		1803-FV -1403		2047-1803-EPR-PID-0014																							
	2	Service	RCSP HOPPER R-104 AERATION PLANT AIR																											
	3	Line Number	Equipment Name		6"-AP-1803-1408-A1A-NI																									
	4	Area Classification	Zone 1 Gr.IIC T3																											
	5	Ambient Temperature:	Min.	Max.	4.4 °C		46.7 °C																							
	6	Allowable Sound Pressure Level	dBA		85 dBA																									
	7	Tightness Requirements	ANSI V																											
	8	Available Air Supply Pressure:	Min.	Max.	3.5 kgf/cm ²		7 kgf/cm ²																							
	9	Power Failure Position	Open																											
	10	Actuator Design Pressure:	kgf/cm ² -g																											
PIPE LINE	11	Line Size and Schedule	Inlet	Outlet	6 in STD		6 in STD																							
	12	Pipe Material	Carbon Steel																											
	13	Pipe Insulation	N/A																											
PROCESS CONDITIONS	14	Process Fluid	PLANT AIR																											
	15	Upstream Condition	Gas/Vapor																											
	16	Differential Pressure	10 kgf/cm ²																											
	17		Units	@ Max. Flow	@ Norm. Flow	@ Min. Flow																								
	18	Flow Rate	Nm ³ /h	2640	2400	2920																								
	19	Inlet Pressure	kgf/cm ² -g	4,3	4,33	4,4																								
	20	Pressure Drop	kgf/cm ²	1,72	1,74	1,83																								
	21	Inlet Temperature	°C	40	40	40																								
	22	Inlet Density / Specific Gravity / Molecular Mass	-		28,96																									
	23	Inlet Compressibility Factor	-	0,999	0,999	0,999																								
	24	Inlet Viscosity	cP	0,0191	0,0191	0,0191																								
	25	Inlet Specific Heats Ratio	-	1,4	1,4	1,4																								
	26	Inlet Vapour Pressure																												
	27	Design Pressure / Temperature	10	kgf/cm ² -g	65	°C																								
	CALCULATED RESULTS	28	Flow Coefficient Cv	-	53,8	48,5	57,6																							
		29	Travel	%																										
	RESULTS	30	Sound Pressure Level @ Maximum Flow:	dBA	89,4	89,5	90,4																							
BODY AND TRIM		31	MFR	Model	VTA	VTA	POSITIONER	56	MFR	Model	VTA	VTA																		
		32	Body Type	Single Seat Globe				57	Signal: Inlet	Outlet	4-20 mA+HAR	3-15 Psi																		
	33	Body Size	Trim Size	4 in	By Vendor	58		Increase Signal Valve:	Close																					
	34	Rated Cv	Characteris.	By Vendor	Linear	59		Cam Characteristic	Direct																					
	35	End Connec. & Rating	4" 300# RF					60	Bypass	Gauges	No	Yes																		
	36	Body Material	C.S.ASTM A216 WCB					61	Position Transmitter	Yes. 4-20mA+Hart																				
	37	Bonnet Type	Material	Bolted Flange	A216 WCB	62	Voltage	24 VDC																						
	38	Flow Direction	FTO (Flow-To-Open)				SOLENOID VALVE	63	MFR	Model	-	-																		
	39	Flow Action To	Open					64	Type	-																				
	40	Lubricator	Isolat. Valve	No	No	65		When De-Enegr.Valve:	-																					
	41	Guiding	No. of Ports	By Vendor	1	66																								
	42	Trim Type	Low Noise Trim				SWITCHES	67	MFR	Model	-	-																		
	43	Rated Travel	By Vendor					68	Type	Quantity	-	-																		
	44	Plug/ Ball/ Disk Material	410 (13Cr)					69	Contacts / Rating	-																				
	45	Seat Material	St Gr6 (CoCrAlloy)					70	Switching Position	-																				
	46	Cage	Stem Material	-	410 (13Cr)	71																								
47	Gasket Material	SPW AISI 316/Graphite				AIR SET	72	MFR	Model	VTA	VTA																			
48	Stud Bolt Materials	A193 Gr B7/A194 Gr 2H					73	Set Pressure	By Vendor																					
49	MFR	Model	VTA	VTA	74		Filter	Gauge	Yes	Yes																				
ACTUATOR	50	Type	Spring Diaphragm				75																							
	51	Size	Area	By Vendor	By Vendor	TESTS	76	Hydro. Pressure	Yes																					
	52	Air Failure Valve:	Open				77	Leakage	Yes																					
	53	Handwheel	Side Mounted				78																							
	54	Bench Range	By Vendor				PURCHASE	79	Manufacturer																					
	55					80		Model																						
					81	Purchase Order Num.																								
				82	Price	Item Number																								
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B	ARSM	22/12/2022	IFA-Issue For Approval																											
A	JOMM	20/06/2022	IFA – Issue For Approval																											
No.	By	Date	Revision																											

Specification Notes:

Process Notes:

1. Continuous operation
2. Reliable Plant Air from Plant Air receiver V-611
3. Minimum Design Metal Temperature = 4.4°C

				INSTRUMENT SPECIFICATION Control Valve		
B	ARSM	22/12/2022	IFA-Issue For Approval		Sheet 2 of 2	
A	JOMM	20/06/2022	IFA – Issue For Approval			
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